STATE OF LIFE

London Youth Rowing

Social value and wellbeing: an economic impact report for the Active Row programme

State of Life

London

Executive summary

- This report covers a four-year period of LYR's Active Row programme. Throughout those four years, Active Row encouraged 11-18-year-olds from over 70 schools across London to take part in indoor and on-water rowing.
- The mission of London Youth Rowing is to encourage disadvantaged young people to be active and improve their physical, social and mental wellbeing.
- London Youth Rowing deliberately works with schools to engage kids who are less active, not already active nor motivated to be active.
- This report starts to evidence whether LYR's Active Row programme has a positive impact on the wellbeing of these kids and what the social and economic value is of the project in line with the latest HM Treasury guidance on wellbeing economics.

Findings

- 44.9% of Active Row participants are in the 30% most deprived areas by indices of multiple deprivation.
- The wellbeing benefits per person are £6,000 per year which is nearly double the £3,000 average benefit for being physically active. This doubling of value is explained by Active Row reaching audiences in more need - inner city, low socioeconomic areas and ethnic minorities
- Social benefits generated by the Active Row programme every year could be over £17 million per year and £69 million over the 4 years of funding.
- We also observe that the higher levels of life satisfaction associated with participation in Active Row are highest for those who have been involved the longest (more than a year) as well as those who have participated the most frequently (once a week or more).
- Furthermore, the data shows that Active Row seems to be quite effective in 'levelling up' and closing the gap for disadvantaged groups of UK society. The wellbeing benefits of rowing participation are evidenced to be considerably higher for free school meal recipients, disabled respondents, and traditionally disadvantaged ethnic minorities.

Limitations

- This study uses practical data collection to evidence a relationship between rowing and wellbeing - which under the right assumptions may achieve level 3 of the Nesta Standards of evidence[1] - by using a comparison group and a duration of participation variable to show impact over time.
- This is a standard of quasi experimental analysis that is generally sufficient for Green and Magenta book compliance but is not perfect.
- Establishing stronger evidence in favour of causality requires a more elaborate research design, such as a randomised trial or at least a longitudinal study. This is often cost prohibitive and impractical.
- Second, this study falls short of performing full social-cost benefit analysis. It involved only one social benefit, although arguably one of the most important - the individual wellbeing impact among young people who take part, and only part of the costs of the programme - the direct funding by sponsors. Investigating other costs (such as volunteer time) and benefits (employment, education, health benefits, crime reduction) is a welcome subject for further research.

[1] https://media.nesta.org.uk/documents/standards_of_evidence.pdf

1. Introduction

The Tideway Tunnel is a 25 km tunnel under the River Thames that improves London's sewerage network and is destined to considerably reduce the number of sewer overflows into the river Thames. It has been under construction since 2016, and is scheduled to be completed in 2024 and become operational in 2025.

Tideway's Legacy Programme consists of a series of commitments to deliver wider social, economic and environmental benefits to society. Among these is a commitment to "Inspire people to engage in river activities and support events that will help people reconnect with the River Thames" (commitment 45).

This objective is set to be achieved by Tideway by means of a partnership with several supporting organisations. One such organisation is London Youth Rowing (LYR). In 2017 Tideway joined forces with Sport England to fund Active Row for a four-year period. Active Row is a programme run by LYR that encourages 11 to 18-year-olds in over 70 schools across London to take part in indoor and on-water rowing.

The initial version of this report was commissioned to evaluate the social impact and value created by this partnership, both to be presented directly to the numerous stakeholders of Active Row, and with a view for the results to feed into the overall social impact evaluation of the Tideway Legacy Programme, which State of Life was commissioned to undertake.

This is a second, updated version of the report, commissioned by London Youth Rowing in an attempt to improve the validity of the findings after collecting more data. The push for extra data was successful and sample size increased more than 4 times compared to the first version of the report. This considerably improved statistical significance and overall robustness of the findings and allowed us to analyse the impact of rowing in more detail.

This study uses a wellbeing approach to estimate social impact as it is defined in the latest version of HM Treasury's Green Book on page 5 where it states '*social value…is based on the principles and ideas of welfare economics and ...therefore includes all significant costs and benefits that affect the welfare and wellbeing of the population*'. The foundation of this approach is the idea that participants' improved personal wellbeing is the main socially desirable outcome of the rowing programme.

The correlation between young people's wellbeing and participation in Active Row is based on data collected via an individual-level quantitative survey of participants and non-participants. Statistical and econometric techniques (regression analysis) are applied to this data. The estimated impact is then converted into monetary terms using the latest UK government Guidance on analysing and monetising wellbeing for policy appraisal and evaluation.

In the remainder of this brief report, we first describe the methodology used to determine the social impact and value of Active Row in more detail. Then we present the results of the survey, wellbeing impact estimation and valuation. This is followed by guidance on how to interpret these results and a short conclusion.

2. Methodology and data

To measure the effect of the Active Row programme on personal wellbeing, State of Life developed a bespoke survey and London Youth Rowing administered it to pupils from the partner schools. A total of 541 responses were received between 24 May 2022 and 4 January 2023, of which 16 were blank or invalid (not within the target age range, obviously non-genuine responses, did not respond to the LYR participation question etc.), and the remaining 525 were valid responses. 243 of these were from Active Row participants and 282 were from non-participants in the same schools (also known as the control group). For research purposes, this is a very good balance between participants and non-participants.

After presenting basic summary statistics for the participant and control groups, we move on to perform multiple linear regression analysis of life satisfaction and other socially desirable outcomes on participation in rowing with all the relevant available demographic information included as control variables. This provides stronger evidence that the relationship between personal wellbeing and Active Row participation is not due to any underlying inherent differences in wellbeing between the two groups (participants and non-participants). However, it still cannot be considered to prove a causal impact of rowing on wellbeing, but it is the most robust approach that is practically feasible given the available data and is more advanced than much of the analysis performed in evaluations of this type of intervention.

We complement these main findings with additional regression analysis that breaks down the effect of rowing into different categories: first - by the nature of involvement (frequency, duration of attending rowing activities, as well as the type of activity attended); second - by

the demographic characteristics of the participants (age, gender, ethnicity, disability, receiving free school meals).

The resulting estimate of the change in life satisfaction associated with participation in rowing (from the main model) is then monetised using the central valuation rate recommended in the latest Wellbeing Supplementary Guidance to the HM Treasury Green Book. The rate is £13,000 / WELLBY. One WELLBY is a change of 1 life satisfaction point on the standard 0-10 scale endorsed by the Office for National Statistics, affecting one person over one year.

Therefore the number of WELLBYs generated by Active Row every year is taken to be the estimated regression coefficient multiplied by the average number of participants in the rowing programme in a typical year. This value is treated as a running benefit, accruing every year for as long as the programme is running. This assumption ignores the fact that a participant's increased wellbeing may persist after they leave the programme, and our value estimate is in this sense conservative.

3. Findings

3.1. Descriptive statistics

First we present summary statistics for the outcomes and demographic composition of the rowing participants and non-participants in the sample.

Table 1. Descriptive statistics

London Youth Rowing participation	**No**	**Yes**	**Total**
Sample Size	282	243	525
MEANS			
Duration of rowing participation (in months)		10.18	
Times rowed during last school term		14.71	
Life satisfaction (0-10)	6.88	7.31	7.08
Happiness (0-10)	6.71	7.19	6.93
Anxiety (0-10)	4.41	4.22	4.32
Worthwhile (0-10)	6.81	7.27	7.03
I've been feeling optimistic about the future (1-5)	3.32	3.5	3.4
I've been feeling useful (1-5)	3.22	3.51	3.36
I've been feeling relaxed (1-5)	3.23	3.45	3.33
I've been dealing with problems well (1-5)	3.25	3.5	3.37

I've been thinking clearly (1-5)	3.54	3.66	3.59
I've been feeling close to other people (1-5)	3.67	3.72	3.69
I've been able to make up my own mind about things	3.7	3.83	3.76
Short WEMWBS total score, 7-35	23.44	24.88	24.11
Resilience (1-4)	3.13	3.34	3.22
Loneliness (1 often/always lonely - 5 never lonely)	3.2	3.23	3.21
Trust (1-4)	2.78	2.91	2.84
Volunteering (1 No, 2 Yes)	1.48	1.59	1.53
General health (1-5)	3.95	3.96	3.96
Days of 30+ minutes of physical activity / week	4.33	4.43	4.38
Days of 60+ minutes of physical activity / week	3.27	3.43	3.34
Motivation for sport/exercise (1-5)	3.28	3.42	3.35
Likely to find a suitable job for you (0-10)	7.2	7.32	7.25
Likely to be successful and get ahead (0-10)	7.31	7.25	7.28
Connection with the river (1-7)	3.03	3.5	3.25
Age (in years)	12.99	13.22	13.1
Local area deprivation, IMD decile (1 - most deprived)	4.35	4.24	4.3
PERCENTAGES			
Gender			
1. Female	52.5	51.9	52.2
2. Male	46.4	46.5	46.4
3. Other	1.1	1.7	1.3
Ethnicity			
1. White British	11.4	18.3	14.6
2. White Other	12.5	16.3	14.2
3. Asian or Asian British	43.6	32.1	38.2
4. Black or Black British	21.6	19.6	20.7
5. Mixed	5.9	11.7	8.6
6. Other	5.1	2.1	3.7
Religion			
1. No religion	14.4	24.2	18.9
2. Christian	31.7	40	35.5
3. Buddhist	1.8	0.4	1.2
4. Hindu	16.9	11.3	14.3
6. Muslim	33.1	22.5	28.2
7. Sikh	1.4	0.8	1.2
8. Other religion	0.7	0.8	0.8
Local area deprivation, 3 categories			

30% highest deprivation	31.1	44.9	37.5
40% medium deprivation	65.5	46.1	56.5
30% lowest deprivation	3.4	9	6
Disability	6.9	12	9.3
Free School Meals	23.9	29.2	26.4
Type of rowing activity			
Indoor rowing at school		91.8	
On water rowing		34.6	
Competition(s) including NJIRC		24.3	
Duration of involvement with LYR, categorical			
Up to 2 months		28.6	
3-6 months		21.9	
7-12 months		25	
More than 12 months		24.5	
Average frequency of involvement, categorical			
Once a month or less		28.5	
More than once a month but less than once a week		38.5	
Once a week or more		33	

Initial findings suggest considerably higher levels of wellbeing among participants, and also a substantive difference in overall mental health. Trust, resilience and volunteering rates are also higher among school rowers.

There is barely any difference in the general health levels of rowers and non-rowers, which means that the higher wellbeing of rowers can not be attributed to higher health levels to begin with. Rowing participants are more physically active - they do more days per week of both 30+ minutes and 60+ minutes of physical activity. They also have higher motivation / enjoyment levels from taking part in exercise and sport.

Finally, school rowers score considerably higher in terms of their self-assessed degree of connection with the River Thames than non-rowers (3.5 vs. 3.0 on average on a 1 to 7 scale), which is encouraging evidence that the Active Row partnership did achieve the stated objective of Tideway Legacy commitment 45, especially after being confirmed by more statistically robust regression analysis further below.

In terms of demographics, participants and non-participants are rather similar in terms of their age and gender composition. There are fewer Asian respondents but more Mixed

ethnicity and White respondents among rowers, although the ethnic composition of the whole sample and both its subgroups is strongly dominated by ethnic minorities (which was confirmed by LYR to reflect the proportion of ethnic minorities among the full population of Active Row). There are also more non-religious individuals and also more people with disabilities among rowers. Rowers also have a slightly higher proportion of free school meals recipients as well as a bigger share living in the 30% most deprived local areas.

3.2. Regression results

We proceed with multiple linear regression analysis. This allows us to ensure that the observed differences in wellbeing and other outcomes are associated with Active Row participation and not a consequence of demographic differences. Results are shown below.

Every row represents a separate multiple linear regression model, where the outcome (dependent variable) is presented in the first column. The second column shows the coefficient of our key treatment variable - which is always Active Row participation (takes the value 1 for participants, 0 otherwise). This means that the coefficient can be interpreted as the average difference in the respective outcome between Active Row participants and non-participants, after taking all differences in the included control variables into account.

Every model also includes the following control variables: age, gender, ethnicity, religion, disability, free school meals, deprivation (IMD 3 categories), and general health (except when it is used as the outcome); their coefficients are not shown to conserve space, but full results are available here: LYR output.xlsx.

Table 2. Regression results by outcome

Outcome variable (answer scale)	Coefficient
Life satisfaction (0-10) **- main result**	0.466***
Happiness (0-10)	0.555***
Anxiety (0-10)	-0.491*
Worthwhile (0-10)	0.619***
Mental health (short WEMWBS 7-35)	1.585***
Resilience (1-4)	0.234***
Loneliness (1-always to 5-never)	0.132
Trust (1-4)	0.118

Volunteering (Yes/no)	0.126***
General health (1-5)	-0.029
Days of 30 minutes of physical activity / week	0.124
Days of 60 minutes of physical activity / week	0.229
Motivation for sport/exercise	0.149**
Likelihood to find a suitable job for you	0.121
Likelihood to be successful and get ahead	0.094
Connection with the river (1-7)	0.552***

Stars indicate statistical significance levels: * p<10%, ** p<5%, *** p<1%.

As we can see, participation in the school rowing programme is associated with statistically significant increases in all of the ONS4 personal wellbeing measures - life satisfaction, happiness, anxiety (although only at a 10% level[2]) and the sense that one's life is worthwhile, and the magnitude of the increase is quite large (larger than some important life events such as marriage although somewhat smaller than the effect of finding a job or improving health).

Other outcomes that increase with rowing participation are mental wellbeing (WEMWBS score), resilience, volunteering, motivation for (enjoyment of) sport and exercise, and reconnection with the River Thames. Other relationships are not statistically significant at the 10% level. There are no statistically significant negative outcomes - the negative coefficient on anxiety is actually a positive relationship because a reduction in anxiety is the desired outcome.

It is worth noting that we have performed a sensitivity check on the extent to which this estimated effect may be due to differences in motivation between rowing and non-rowing respondents in our sample. Other work (e.g. forthcoming report by State of Life for the Youth Sport Trust on the impacts of sport in school) shows that differences in motivation for physical exercise are responsible for around 40% of the observed relationship between physical activity in school and life satisfaction. Fortunately, our survey contains a 'motivation for sport/exercise' question[3], which we can include as an additional control variable in the regression to test whether this hypothesis also holds for rowing.

[2] The statistical significance level represents the likelihood that this result is observed under the assumptions that one is randomly sampling from a population where there is no true relationship between the variables in question (e.g. no impact of rowing involvement on life satisfaction).

[3] Thinking about sport/exercise in general, how much do you agree or disagree with this statement: - I enjoy taking part in exercise and sports? 4. Strongly agree 3. Agree 2. Disagree 1. Strongly disagree

Including motivation as a control leaves the main result almost unchanged - the coefficient of Active Row participation on life satisfaction goes down from 0.466 to 0.448, or less than 4%. Coefficients on other outcomes reduce by no more than 10%. This means that motivation does not cause much omitted variable bias in this study, and the main results have withstood this sensitivity check. This is not surprising as the difference in average motivation levels between participants and non-participants (see table 1) is not very big.

3.2.1. Split by type of involvement

The increased sample of the survey allows us to obtain statistically significant results while differentiating between different types of rowing involvement. In Table 2B below we report the coefficients from a few more regression models where we distinguish between different duration, intensities, and activities of rowing involvement.

All coefficients are to be interpreted as **differences in life satisfaction relative to those who do not participate in rowing at all**. The same control variables as before were used.

Table 2B. Regression results disaggregated by duration, frequency, activity type

Treatment variable subcategory (rowing type)	**Coefficient**
Base model (all rowing participants)	0.466***
By duration of participation:	
Up to 2 months	0.047
3-6 months	0.656**
7-12 months	0.509*
More than 12 months	0.862***
By frequency of participation:	
Once a month or less	0.687**
More than once a month but less than once a week	0.009
Once a week or more	1.209***
By activity type:	
Indoor rowing at school	0.476***
On water rowing	0.519**
Competition(s) including NJIRC	0.512*

Stars indicate statistical significance levels: * p<10%, ** p<5%, *** p<1%.

The main take-home message from this analysis is that those who have been involved the longest, as well as the most frequently, experience the highest wellbeing levels relative to non-participants. It would seem that the more you participate in school rowing, the better for you. However, the pattern shown by the evidence is not as neat - with there being a dip in the second highest category in both instances. The sample size is not sufficient to tell whether this is merely by chance or an actual picture of how the wellbeing benefits of rowing change over time (e.g. they could increase with initial enthusiasm, then cool off after a period of settling in and fatigue, and then experience a resurgence for those who make it through and stay the longest).

In terms of the different types of rowing - indoor, on water, competitions - they all seem to show relatively stable positive relationships with wellbeing across the board, with negligible differences among them. The difference in significance levels is tied to the difference in sample sizes - only 59 participate in competitions vs. 84 on water rowing and 223 indoor.

3.2.2. Split by demographics of participants

Similarly to the above, we look at how the life satisfaction benefits associated with school rowing vary for different subgroups in the sample - males and females, ethnic groups, children from lower socioeconomic backgrounds etc. Just as above, we only look at life satisfaction to conserve space, as this is our main outcome also used for valuation.
The results below were derived using regression models with interaction terms[4] between Active Row participation and the respective demographic variables. The same control variables as always have been used but not reported here, with full results in LYR output.xlsx .

[4]From a technical perspective, an interaction term is a product of two indicator variables.

Table 2C. Regression results disaggregated by demographic subgroups

Sample subgroup	Coefficient
Base model (all participants)	0.466***
By gender:	
Female	0.428*
Male	0.482*
By broad ethnic group:	
White British	0.212
White Other	0.143
Asian or Asian British	0.667**
Black or Black British	0.843**
Mixed	0.082
By disability:	
No	0.432**
Yes	0.844
By free school meals:	
No FSM	0.343
FSM	1.060***

Stars indicate statistical significance levels: * p<10%, ** p<5%, *** p<1%.

In Table 2C above we can see good evidence that the Active Row programme can play a positive role in 'levelling up,' that is, closing gaps (in personal wellbeing in this case) between different groups in UK society. The benefits associated with participation in this school rowing programme are about three times higher for respondents on free school meals, and also about twice as big for those who have a disability (though not statistically significant). Furthermore, across all ethnic subcategories in the sample, the highest benefits are experienced by Asian and Black respondents, two ethnicities not traditionally associated with rowing.

3.3. Wellbeing valuation

The coefficient of rowing participation on life satisfaction from table 2 earlier - 0.466 - forms the basis for our impact measurement and valuation of the overall Active Row programme.

This is the extra life satisfaction that school rowers experience on average compared to their peers who are not involved in rowing. We assume that this is the estimated effect of rowing on an individual's wellbeing, even though we cannot establish causality with our current research design. The regression analysis accounts for key observable demographic factors that we know influence wellbeing outcomes and is considerably more robust than most other approaches in comparable research. Nevertheless, it is not possible to exclude the possibility of unobservable factors driving the difference between the two groups.

London Youth Rowing supplied us with administrative data on the number of participants in the Active Row programme. We discard the numbers from the more recent years, when rowing activities were either completely suspended or reduced because of COVID19. Therefore we assume the number of typical participants in a year to be an average of participation levels in 2017/18, 2018/19 and 2019/20 - 2866 per year (see Table 3 below).

This means that the total wellbeing impact of the programme in any given year is around 1,336 WELLBYs (the product of the two numbers above). This is assuming that any wellbeing increases last as long as the individual is participating in Active Row and does not take into account any residual or long-term improvements in wellbeing after participation ends.

Applying the valuation rate of £13,000 per WELLBY, recommended by the Green Book supplementary guidance for social cost-benefit analysis, it would result that the **social benefits generated by the Active Row programme every year could be over £17 million** (and the benefit per person per year is around £6,000).

This is a positive, but at the same time plausible finding. Wellbeing analysis using large national panel data sets (compiled in State of Life's WELLBY Value Guide) shows that regularly engaging in sport and/or physical activity is associated with a positive wellbeing impact worth up to £3,000 per person per year. At the same time, previous studies have shown that programmes working with lower socioeconomic groups or other groups with prior disadvantage in wellbeing can be 2-3 times more effective than for the average population.

3.4. Social cost-benefit analysis

This report covers a period of 4 years (2017/18, 2018/19, 2019/20 and 2021/22). This means that the total wellbeing benefit over this time frame is just over £69 million.

In this period, Tideway allocated £404,000 to London Youth Rowing to fund Active Row. Sport England allocated another £1,320,000, arriving at a total cost of £1,724,000. This means that **the benefit-cost ratio of the Active Row programme is 40.3, or over £40 in social value returned for every £1 invested in the programme**.

Table 3. Wellbeing valuation and social cost-benefit analysis (CBA)

2019/20 participants	2880
2018/19 participants	2936
2017/18 participants	2783
Average participants	**2866**
Wellbeing impact	0.466
Total WELLBYs	**1335.7**
Value per WELLBY	£13,000
Total yearly value	**£17,364,247**
Number of years	4
Total value over time	**£69,456,989**
Tideway funding	£404,000
Sport England funding	£1,320,000
Total funding	**£1,724,000**
BCR	**40.29**

One might wish to discount the total benefits to be more conservative, because the wellbeing estimate may be subject to selection or omitted variable bias, and therefore may be an overstatement (see next section for details). Even if one were to only consider 25% of the estimated wellbeing effect to be true, the social value over time of the Active Row programme would still be over £17 million and the return on investment ratio (BCR) over 10 to 1.

4. Interpretation and Conclusion

This study is a basic attempt to assess the wellbeing impact of the Active Row programme and its corresponding social value. It does so by analysing the responses from a survey of 525 young people aged 11 to 18, of which almost half participate in rowing activities and the remainder do not.

The data shows that rowing participants score higher for personal wellbeing and some other outcomes than non-participants. This positive association remains when we control for

observable demographic characteristics in multivariate regression analysis, and measures 0.466 points on the 0-10 life satisfaction scale.

We also observe that the higher levels of life satisfaction associated with participation in Active Row are highest for those who have been involved the longest (more than a year) as well as those who have participated the most frequently (once a week or more). This may imply that rowing is a social good of which more is better - a positive finding for all stakeholders of the rowing programme and a good argument in favour of increasing the programme's scale and scope.

Furthermore, the data shows that Active Row seems to be quite effective in 'levelling up' and closing the gap for disadvantaged groups of UK society. The wellbeing benefits of rowing participation are evidenced to be considerably higher for free school meal recipients, disabled respondents, and traditionally disadvantaged ethnic minorities. Therefore programmes such as Active Row may be an effective tool to bring their welfare levels closer to the traditionally more privileged groups.

Given that Active Row has had an average of over 2800 participants between 2017/18 and 2019/20, every typical year of the programme brings about a positive wellbeing impact of over 1300 WELLBYs to UK society. Using the recommended valuation rate in the Green Book supplementary guidance on wellbeing, this is equivalent to over £17 million in social benefits generated by Active Row every year.

For the four-year period covered in this report, the total wellbeing benefits would be £69 million. The total funding for Active Row over this time period was £1.7 million, split between Tideway and Sport England. This results in a 'bottom line' benefit-cost ratio metric (BCR, also known as SROI) of around 40 - that is a return of £40 in social value for every £1 invested.

4.1. Limitations of the study

There are several limitations to this study. The first is that the relationship between Active Row participation and wellbeing cannot be seen as causal. Establishing stronger evidence in favour of causality requires a more elaborate research design, such as a randomised trial or at least a longitudinal study. It is therefore uncertain whether the observed regression coefficient represents the true impact of the programme. Reasons which prevent us from claiming causality may include selection bias. Table 1 shows the observable demographic

differences between participants and non-participants that we control for, but there also may be unobservable differences (e.g. personality traits) which our model is unable to capture.

Second, this study falls short of performing full social-cost benefit analysis. It involved only one social benefit, although arguably one of the most important - the individual wellbeing impact on young people who take part, and only part of the costs of the programme - the direct funding by sponsors. Investigating other costs (such as volunteer time) and benefits (employment, education, health, crime reduction) is a welcome subject for further research.